C64uRemote for the M5StickC Plus2

User Manual

Version 1.0

Contents

Welcome

C64uRemote turns an **M5StickC Plus2** into a pocket-sized remote control for the **Commodore 64 Ultimate (c64u)** and the **Ultimate64 Elite-II**. Over Wi-Fi it drives the machine: reset, reboot, power off, open the Ultimate menu and change the CPU speed.

The optional **Unit RFID2** on the Grove port adds the thing that makes the little stick genuinely practical: you set up its **Wi-Fi with an NFC card**. A card written on the M5Dial or the M5Stack Core goes onto the reader, and the stick takes over network and password — it never has to go back to a computer. The same **command cards** as on the other devices work here too, and the stick can write cards itself.

What it cannot do is start games from a card — it has no memory card. See the *NFC cards* chapter.

This project builds on the original idea by Karl Prosser (@klumsy) and was extended for the M5StickC Plus2 by Martin Oswald (@mad, 1MHz.de).

What you need

Required:

- An M5StickC Plus2
- A Commodore 64 Ultimate or Ultimate64 Elite-II on the same Wi-Fi

Optional, for NFC cards:

- A **Unit RFID2** (WS1850S) on the Grove port (HY2.0-4P)
- NFC cards: NTAG215 recommended, NTAG213/216 and MIFARE Classic also work

Optional, for input:

- A **Hat MiniJoyC** on the HAT port

Both add-ons are detected at power-up. If one is missing, everything else keeps working. RFID2 and MiniJoyC sit on separate connectors and do not interfere, so you can use both at once.

First setup

Before the stick can find your C64, the Wi-Fi credentials and the address of the C64 have to be stored once. There are two ways to do that, and neither needs **reflashing**:

1. Present a **Wi-Fi card** (requires the RFID2)
2. Start the **setup portal** and fill in a form in your browser

Both are covered in the *Setting up Wi-Fi* chapter. The data goes into internal storage and survives every restart.

If you prefer to supply the credentials at build time, put them into `build_env.h` (see the technical documentation). They then act as the initial values for the very first start and are overridden by anything you set on the device later.

On the first start with no network stored the stick reports **WLAN EINRICHTEN** (set up Wi-Fi). If the network is known but the C64 address is missing, it reports **HOST FEHLT** (host missing).

The MiniJoyC status LED

With a MiniJoyC attached, its RGB LED shows the overall state without opening any menu:

LED	Meaning
green	connected, C64 answers, password accepted
blue	Wi-Fi up, but the C64 does not answer
blue, blinking	C64 reachable, but the password is wrong
red	no Wi-Fi connection

Brightness and on/off are in *Settings*.

Operation

The M5StickC Plus2 has three buttons: the **large button on the front** (A), the **small button on the side** (B) and the **power button** on the left.

Button	On the home screen	In lists and menus
A (large, front)	Reset. Twice in quick succession = reboot	One level back
B (side)	Open the Ultimate menu	Next entry
Power (short)	Open the menu	Select / execute

Holding the power button switches the stick itself off — that is a hardware function and has nothing to do with the C64.

With a MiniJoyC

With a MiniJoyC attached it gets more comfortable:

Input	On the home screen	In lists and menus
Left	–	One level back
Right	Open the menu	Select / execute
Up	Reset	Previous entry
Down	Power off (asks first)	Next entry
Press stick	Ultimate menu	Ultimate menu

If the MiniJoyC is unplugged while running, the stick reports *MINIJOYC OFFLINE* and carries on with the buttons.

The home screen

The home screen shows the 1MHz logo — either calm or with changing effects (*Water*, *RotoZoom*, *SineWave*, *Ripple*, *Raster*). Which effects run and for how long is set in *Settings*.

Unless *Auto-NFC* is *Off*, the stick polls the card reader in the background here. A card is picked up straight away, without going into any menu first — the most convenient way in daily use.

The menu

A press of the power button (or joystick right) opens the menu:

Entry	What it does
PowerOff	switch the C64 off — asks once
CPU Speed	pick a clock speed, the list comes from the c64u
NFC / RFID	read and write cards
WLAN	set up and manage networks
Joystick Swap	swap the joystick ports on the C64 (Normal ↔ Swapped)
Status	Wi-Fi, reachability, password, CPU, address
Settings	all settings

PowerOff shows *POWER OFF? NOCHMAL!* — a second press within two seconds really switches off, anything else cancels.

Changing the CPU speed

CPU Speed shows the current value at the top and below it the list of steps your c64u actually offers. The stick queries them when the screen opens; if it is not connected at that moment, it shows a fallback list.

Select and execute — the new value is shown briefly and taken over at the top.

Swapping the joystick ports

Some games expect the joystick in port 1, others in port 2. Instead of moving the cable, the mapping can be swapped inside the C64.

Choose **Joystick Swap** from the menu — every press toggles between *Normal* and *Swapped*, and *JOY Swapped* or *JOY Normal* appears briefly.

Settings → c64u Joystick shows the current state and steps through every value your C64 offers: besides *Normal* and *Swapped* there may be *WASD P1* and *WASD P2*, depending on the firmware — the keyboard then drives that port.

This is the mapping **inside the C64**, not the MiniJoyC on the stick. The Ultimate firmware has no dedicated remote command for it; the stick sets the *Joystick Swapper* item in the C64 configuration, exactly like the clock speed. The state therefore survives until it is changed again.

Setting up Wi-Fi

The stick remembers **up to four networks**. At power-up it scans briefly and connects to the known network with the best signal. If that fails, it works through the remaining ones while running.

Way 1: present an NFC card

This is the quickest route, and the actual reason for putting an RFID2 on the stick.

A Wi-Fi card is most easily created on the **M5Dial** or the **M5Stack Core** (there: *WLAN → Auf NFC-Karte*). Hold that card against the stick — done.

Two options:

- **On the home screen**, if *Auto-NFC* is on: simply present the card.
- Via *NFC / RFID → Karte lesen* or *WLAN → Von NFC-Karte*.

The stick takes over network and password, stores both and connects immediately. It waits up to eight seconds for the result and reports *WLAN AKTIV* together with its IP address, or *NETZ NICHT DA*. If it is already connected to exactly that network it reports *SCHON VERBUNDEN* and leaves everything alone — so a card left lying on the reader does no harm.

Way 2: the setup portal

Without a card it works through a private access point:

WLAN → Setup-Portal

The stick then shows:

Network	`C64uRemote-Setup`
Password	`c64ultimate`
Browser	`192.168.4.1`

Join that network with a phone or laptop, open the address in a browser and fill in the form: Wi-Fi name, Wi-Fi password, the address of the c64u and — if set — its password. After *Speichern* (save) the stick shuts the access point down and connects to the new network. The browser connection breaking off in the process is normal.

A password field left empty keeps the previous value — so you can change the C64 address without retyping the Wi-Fi password.

If the portal is not used for five minutes it closes by itself. A press of **A** closes it immediately.

Managing networks

Entry	What it does
Gespeichert	list of all networks. Selecting one connects to it
Auf NFC-Karte	writes the most recently used credentials to a card
Netz loeschen	drop a single network from the list
Alle loeschen	clear the whole list

In the list, *aktiv* marks the network the stick is currently connected to. When a fifth network is added, the least recently used one drops out.

NFC cards

Everything about cards lives under *NFC / RFID*:

Entry	What it does
Karte lesen	present a card, its content is executed
CMD-Karte	pick a command and write it to a card
WLAN auf Karte	write the most recently used Wi-Fi credentials to a card

Without a reader all three show *kein NFC* and a press reports *KEIN RFID2*.

Reading a card

The stick works out for itself what is on the card:

- **Wi-Fi card** → take over the network and connect
- **Command card** → run the command right away
- **Game card** → the notice *BRAUCHT SD-KARTE* (needs an SD card)

The most comfortable way is to switch *Auto-NFC* on and present the card on the home screen.

Command cards

Instead of a game, a card can carry a command for the C64. Such a card is surprisingly handy in daily use: one card for “reset”, one for “Ultimate menu”, one for “off”.

Available commands:

Command	Effect
Reset	reset the C64
Reboot	cold start
Ultimate Menu	open the Ultimate menu
PowerOff direkt	switch off immediately, no prompt
PowerOff with prompt	switch off, but only after confirmation
CPU x MHz	set a clock speed
Swap Joystick	toggle the joystick ports (Normal ↔ Swapped)

Command	Effect
Joystick Normal / Swapped / WASD P1 / WASD P2	set the port mapping to a fixed value

Creating a command card

1. Choose *NFC / RFID → CMD-Karte*.
2. Pick the command from the list. The stick fetches the joystick mappings and the CPU steps from the c64u beforehand; *JOY* or *CPU* is shown on the right.
3. Present a card. *KARTE OK* means: written and read back for verification.

A card can be rewritten as often as you like. The content stays a plain NDEF text record — any NFC app on a phone can read and display such a card.

PowerOff with a prompt

A card that switches the C64 off immediately can be unpleasant if it ends up on the reader by accident. Hence the variant with a prompt:

- Present the card → *POWER OFF? NOCHMAL!*
- Within the time window **present the same card again** or press a button → the C64 goes off
- Do nothing → nothing happens

The length of the window is set in *Settings → NFC-Cmd PowOff* (3, 5, 8 or 15 seconds, 8 s by default). The time is written into the card when it is created.

What is on the card

For anyone who would rather write cards with a phone app: the text is plainly readable.

```
CMD:RESET
CMD:REBOOT
CMD:MENU
CMD:POWEROFF=0      switch off immediately
CMD:POWEROFF=8      ask first, 8 s to confirm
CMD:CPU=10          set the CPU to 10 MHz
CMD:JOY             toggle the joystick ports
CMD:JOY=SWAPPED     set the ports fixed; also NORMAL, WASD1, WASD2
```

Case and spaces do not matter. The M5Dial and M5Stack Core use the same format — one card works on every device.

Wi-Fi cards use the scheme from the Wi-Fi QR code:

```
WIFI:S:MyNetwork;T:WPA;P:secret;;
```

Why there are no game cards

A game card holds the **path of a program file**. To turn that into a running game, the device has to fetch the file from its own microSD card and send it to the C64. The M5StickC Plus2 has no card slot — so it reports *BRAUCHT SD-KARTE* for such a card.

The card itself is left completely untouched and stays valid; it belongs on the M5Dial or the M5Stack Core. The other way round: command and Wi-Fi cards behave identically on all four devices.

Settings at a glance

Every setting is saved immediately and survives a restart.

NFC

Entry	Choices
Auto-NFC	background polling interval on the home screen: *Off*, *1.5s*, *0.7s*, *0.3s*
NFC-Cmd PowOff	default prompt time for a PowerOff command card: 3, 5, 8, 15 s

A shorter interval reacts faster but draws slightly more power. *Off* disables background polling entirely; cards then only work through *NFC / RFID → Karte lesen*.

Display

Entry	Choices
Animations	turn home screen effects on or off
Effect	*Auto* or one fixed effect
Anim Speed	*Slow*, *Normal*, *Fast*
Effect Time	how long an effect runs
Static Time	how long the calm logo sits in between
Brightness	display brightness

MiniJoyC

Entry	Choices
Joy Overlay	small readout of the joystick values on/off
Joy LED	status LED on/off
Joy LED Bright	LED brightness
Joy Threshold	how far the stick must move for a direction to count (40–100)

Other

Entry	Choices
c64u Joystick	port mapping in the C64: *Normal*, *Swapped*, and depending on firmware *WASD P1 / WASD P2*
Factory Reset	all settings back to their defaults

Factory Reset only touches the settings. Stored Wi-Fi credentials survive — those are cleared under *WLAN → Alle loeschen*.

Frequently asked questions

Now and then *Not reached* or *Not verified* shows up and clears again. The HTTP server in the c64u occasionally refuses a connection ("connection refused") although network and address are fine – this happens even with only a single device on the network. Since v1.2.1 the firmware retries a refused call by itself after a short pause, so you will usually not notice. If it stays that way, restarting the c64u helps.

The screen stays dark. That is almost always the board setting at build time, not the hardware. See the technical documentation, chapter *Build environment*.

The stick does not find the C64. Check *Status*. *Disconnected* means no Wi-Fi. *Not reached* means the address is wrong or the C64 is off. *Auth failed* means the c64u password is wrong.

After presenting a Wi-Fi card it says *NETZ NICHT DA*. The credentials are stored anyway. The stick keeps trying in the background; often the network is simply slower than the eight second wait.

A card is not recognised. Move the card a little — the RFID2 antenna sits centred under the case. If *NFC / RFID* shows *kein NFC* everywhere, the reader was not found at startup: check the plug and restart the stick.

The same card keeps triggering over and over. It does not: a card left on the reader is only accepted again after two and a half seconds. Only the prompt of a PowerOff card may be confirmed straight away.

Can I create cards on the stick and use them on the M5Dial? Yes. Command and Wi-Fi cards are interchangeable between all four devices.

License

How this was made

The port, the extensions and these manuals were written with the help of Claude (Anthropic). Concept, idea, hardware decisions and every test on real devices: Martin Oswald (@mad).